2201651-Expriment 8 : Hypothesis Testing

1. What is 2-tailed and 1-tailed T-Test?

- 1-tailed T-Test: Used when the hypothesis tests for a specific direction (greater than or less than).
  Example: Testing if Method A gives higher scores than Method B.
- 2-tailed T-Test: Used when the hypothesis checks for any difference, regardless of direction.
  Example: Testing if there's any difference in scores between Method A and Method B (either higher or lower).
  In short, 1-tailed tests are directional, 2-tailed tests are non-directional.

2. Explain what is Type I and Type II errors with an example.

- Type I Error (False Positive): Rejecting the null hypothesis when it is actually true.
  Example: A medical test wrongly says a person has a disease when they don't.
- Type II Error (False Negative): Failing to reject the null hypothesis when it is false.
  Example: A medical test fails to detect a disease when the person actually has it.

3. What is power of a statistical test?
The power of a test is the probability that it correctly rejects a false null hypothesis.

- It measures a test's ability to detect an actual effect.
- Power = 1 – Probability of Type II Error (β)
- Higher power means a greater chance of detecting a real difference when it exists.
  Typically, a power of 0.8 or 80% is considered acceptable.

4. Explain input and output characteristics for stats.ttest_rel function in your own words:
The stats.ttest_rel() function from scipy.stats is used for paired sample t-tests, comparing two related groups (e.g., before and after treatment).

- Input: Two arrays or lists of the same length, representing related samples.
  Example: stats.ttest_rel(before_scores, after_scores)
- Output: A tuple containing:
- t-statistic: Indicates how much the sample means differ relative to variation.
- p-value: Helps determine if the difference is statistically significant.

5. What is function in stats module for identifying power of a statistical test? Explain output range of it and its significance:
The function used to compute the power of a statistical test in statsmodels is typically:
statsmodels.stats.power.TTestPower().power() for t-tests.

- Output: A value between 0 and 1, representing the probability of correctly rejecting a false null hypothesis.
- Significance:
- Closer to 1 = High power = Low chance of Type II error
- Closer to 0 = Low power = High chance of missing a true effect
- Researchers aim for power ≥ 0.8 to ensure reliable results.